3 The Fermi-Dirac distribution function:

$$f(\epsilon) = \frac{1}{\exp(\frac{\epsilon-\mu}{k_B T}) + 1}$$

indicates the occupation probability of a fermionic state with energy ϵ at the temperature T and chemical potential μ. $k_B = 8.6173324 \times 10^{-5}$ eV/K is the Boltzmann constant. Plot the probability $p(\epsilon) = 1 - f(\epsilon)$ that a state is *unoccupied* for $T = 10$, 300 and 1000 K and $\mu = -1$ eV, for energies ϵ between -3 and 0 eV. Now calculate explicitly $p(\epsilon)$ for energies $\epsilon = 0.0, -0.2, \ldots, -2.8, -3.0$ eV, $T = 300$ K, and the same value of μ. You should have numerical problems for $\epsilon \leq -2.0$ eV and cannot distinguish $p(\epsilon)$ from zero. Look at the formula for $p(\epsilon)$ again and reshape it by bringing the right-hand side to a common denominator and simplifying the numerator.

Expected output. Plot of $p_1 = 1 - f$ for the three temperatures and the energy range indicated. Table with the values of ϵ, p_1 and p_2, where $p_1 = 1 - f$ and p_2 is the reformulated expression (the formatting of the table is not important as long as the results are clear). Answer to the question: why do you get a different result?

energy (eV)	$p_1 = 1 - f$	p_2
0.00	1.00e+00	1.00e+00
-0.20	1.00e+00	1.00e+00
-0.40	1.00e+00	1.00e+00
-0.60	1.00e+00	1.00e+00
-0.80	1.00e+00	1.00e+00
-1.00	5.00e-01	5.00e-01
-1.20	4.36e-04	4.36e-04
-1.40	1.91e-07	1.91e-07
-1.60	8.33e-11	8.33e-11
-1.80	3.64e-14	3.64e-14
-2.00	0.00e+00	1.59e-17
-2.20	0.00e+00	6.93e-21
-2.40	0.00e+00	3.03e-24
-2.60	0.00e+00	1.32e-27
-2.80	0.00e+00	5.77e-31
-3.00	0.00e+00	2.52e-34

4 Given the system of equations $Vx = b$ with the 6×6 Vandermonde matrix $(V)_{ij} := a_i^{j-1}$, with $i, j = 1, \ldots, 6$, and the vectors $a = \left(\frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10}, \frac{5}{10}, \frac{6}{10}\right)$, $b = \left(1, 2, 3, 4, 5, 6\right)^T$:

- Solve the equation system either with SciPy's or NumPy's function `linalg.solve` or with your function from Exercise 2 (if you do so, please copy it in your current directory).
- Now solve the system $Vx_s = b_s$ for the slightly perturbed vector:

$$b_s = b + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0.001 \\ 0 \\ 0 \end{pmatrix}.$$

- Compare the relative error of the two solutions $\Delta x = ||x_s - x||_2/||x||_2$ with the relative error $\Delta b = ||b_s - b||_2/||b||_2$ ($||\cdot||_2$ is the Euclidean norm). From these two errors, calculate a lower bound κ_m for the condition number of the system.
- Calculate the condition number κ of the system of equations using SciPy's or NumPy's function `linalg.norm` and compare it with your estimate. Note: use the 2-norm.

Expected output. The values of Δx, Δb, κ_m, and κ.